

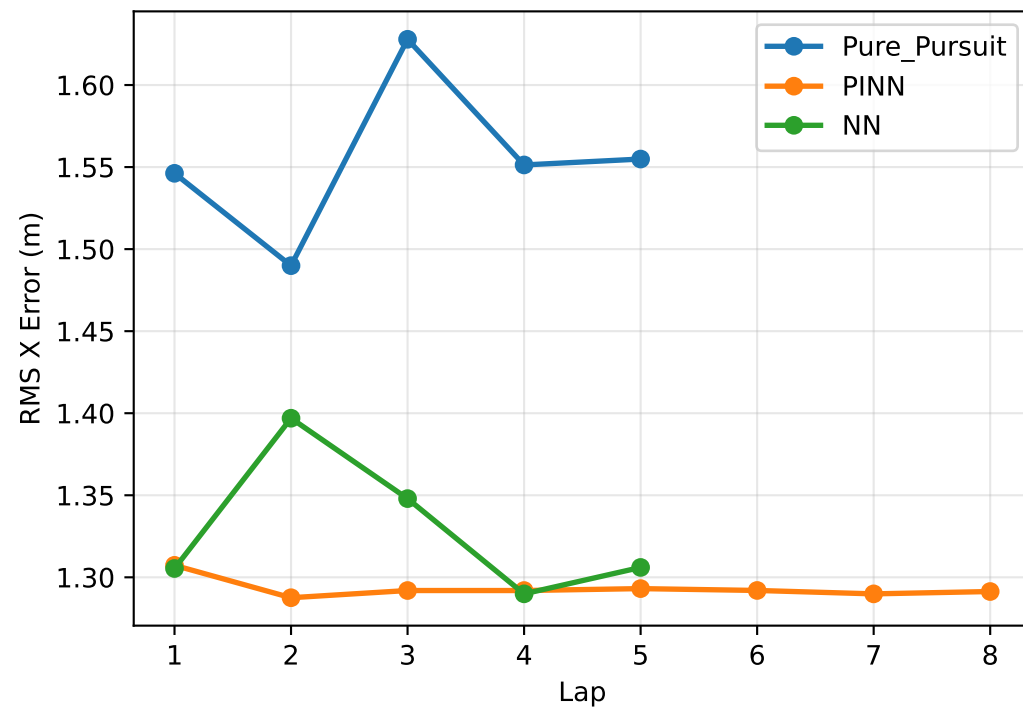
F1TENTH Controller Performance Comparison Report

Generated: 2026-03-17 14:08:13

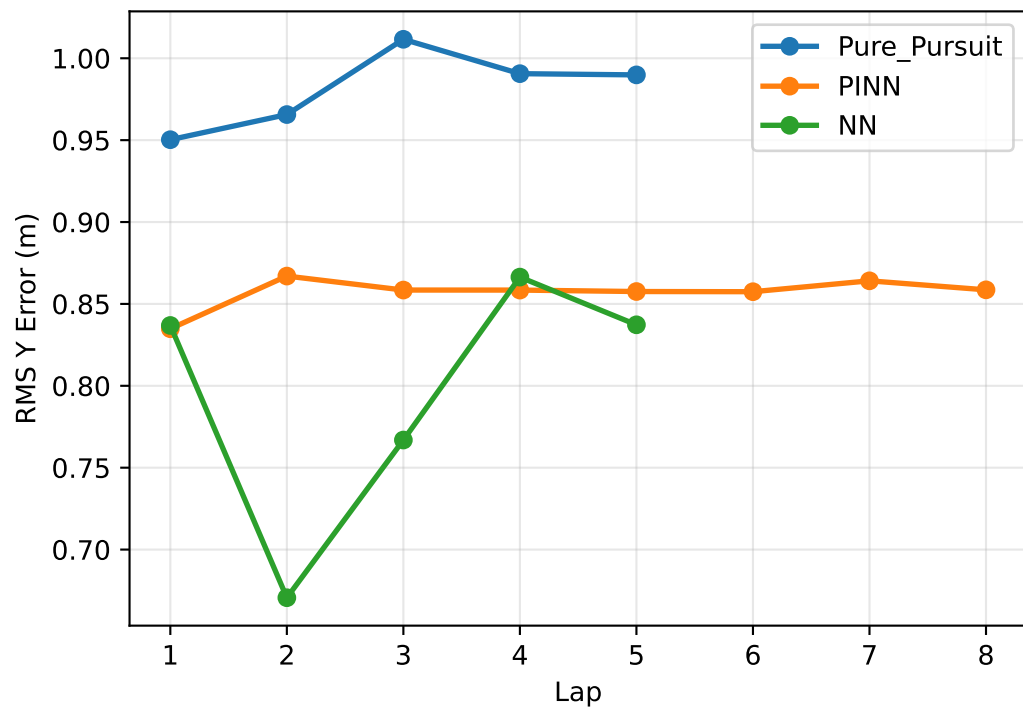
Controller	Laps	Avg Lap Time (s)	Avg RMS X Error (m)	Avg RMS Y Error (m)	Avg RMS Pos Error (m)	Avg CTE (m)	Max CTE (m)	Steering Smooth	Avg Correlation
Pure_Pursuit	5	9.187	1.5540	0.9816	1.8381	0.1439	0.3451	0.3473	0.5271
PINN	8	8.408	1.2931	0.8571	1.5514	0.1907	0.5955	0.3481	0.5521
NN	5	11.933	1.3292	0.7956	1.5512	0.2614	0.6121	0.2908	0.5607

Positional Error Analysis

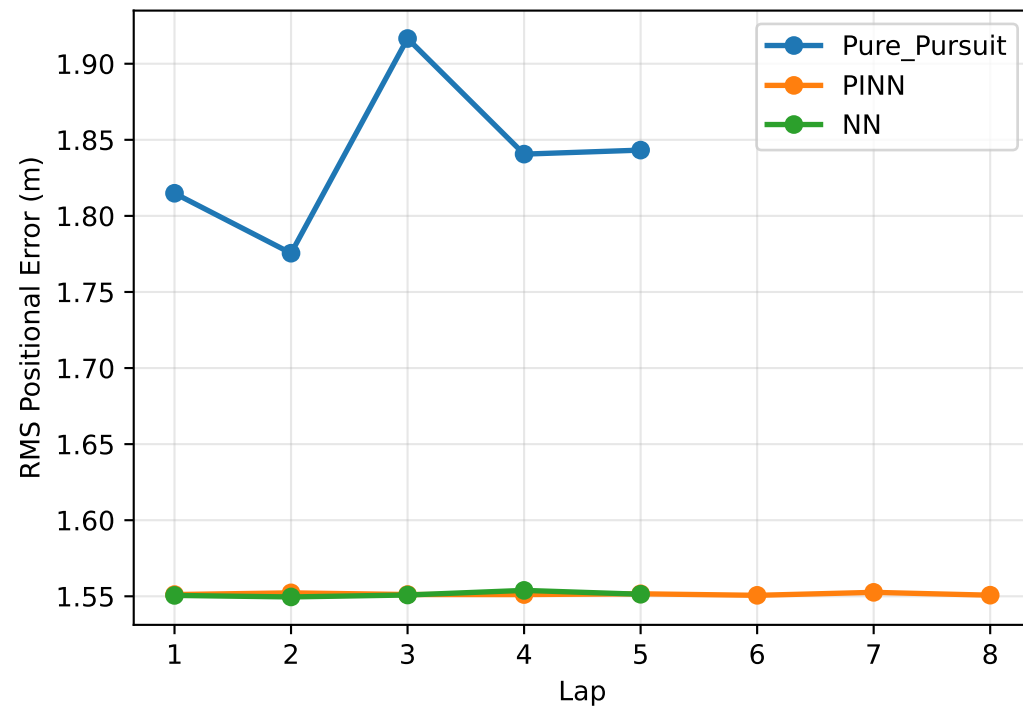
RMS X Error (m)



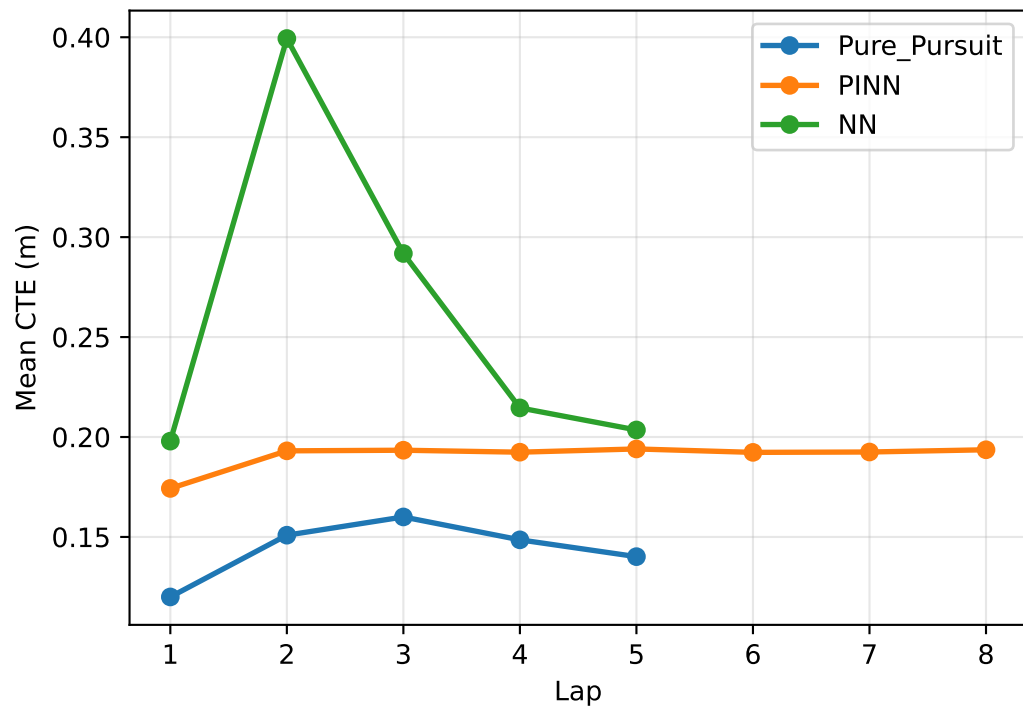
RMS Y Error (m)



RMS Positional Error (m)

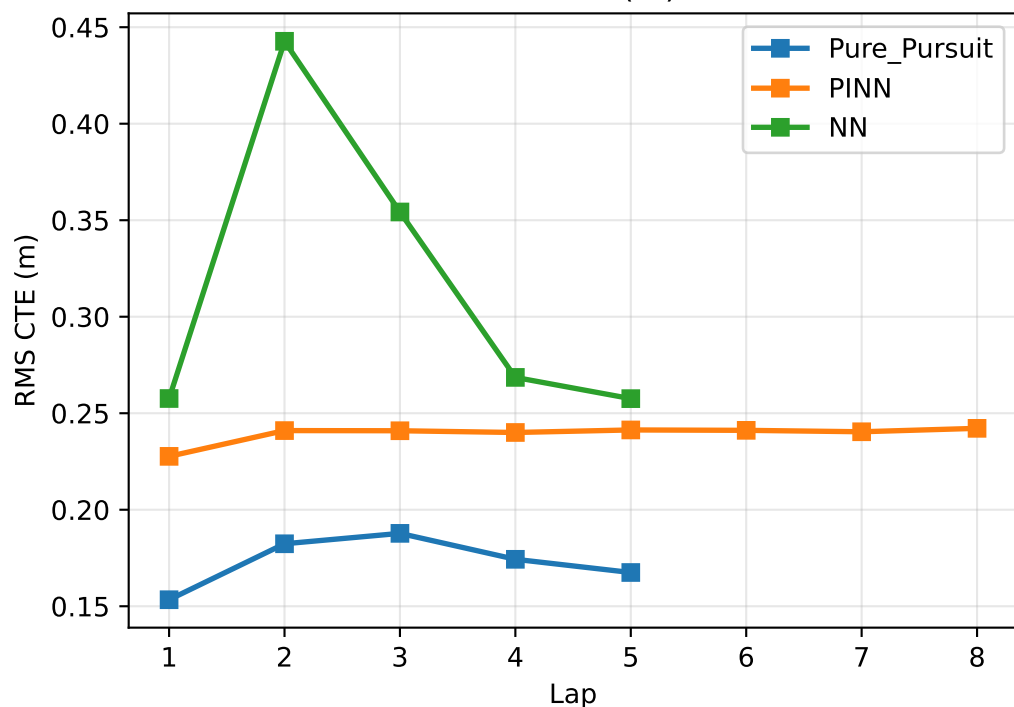


Mean CTE (m)

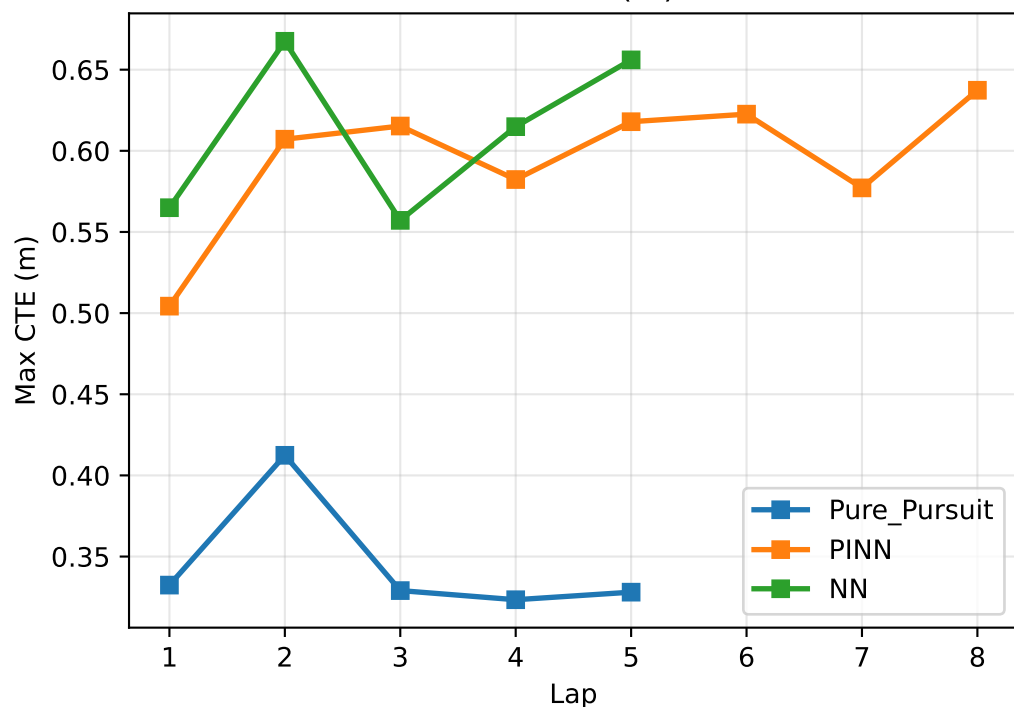


Cross-Track Error & Smoothness Analysis

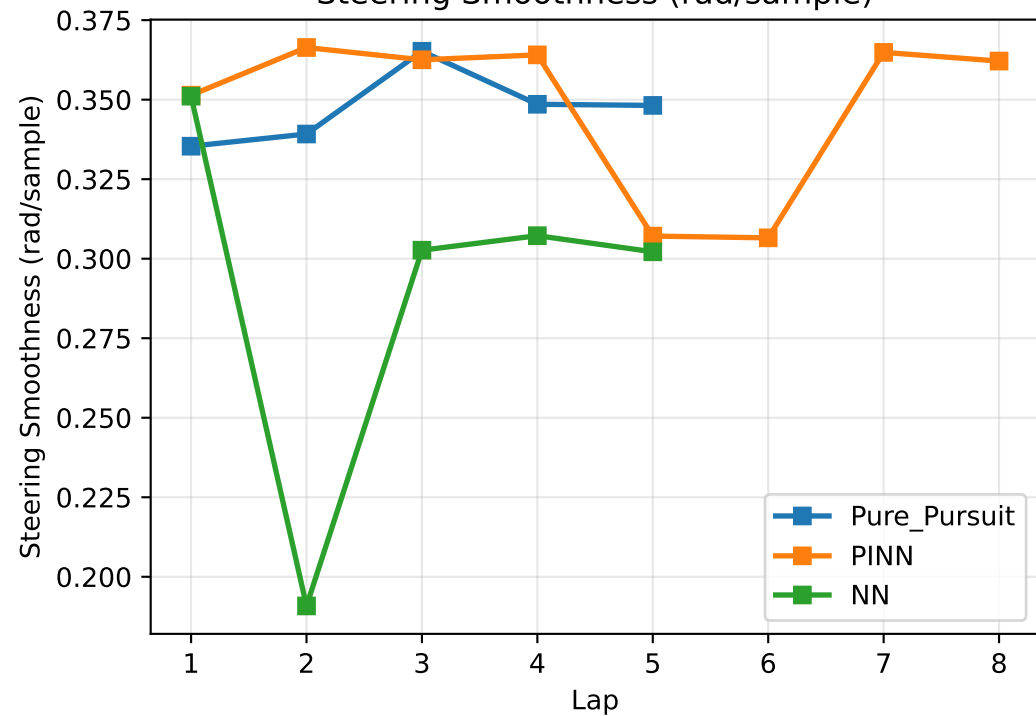
RMS CTE (m)



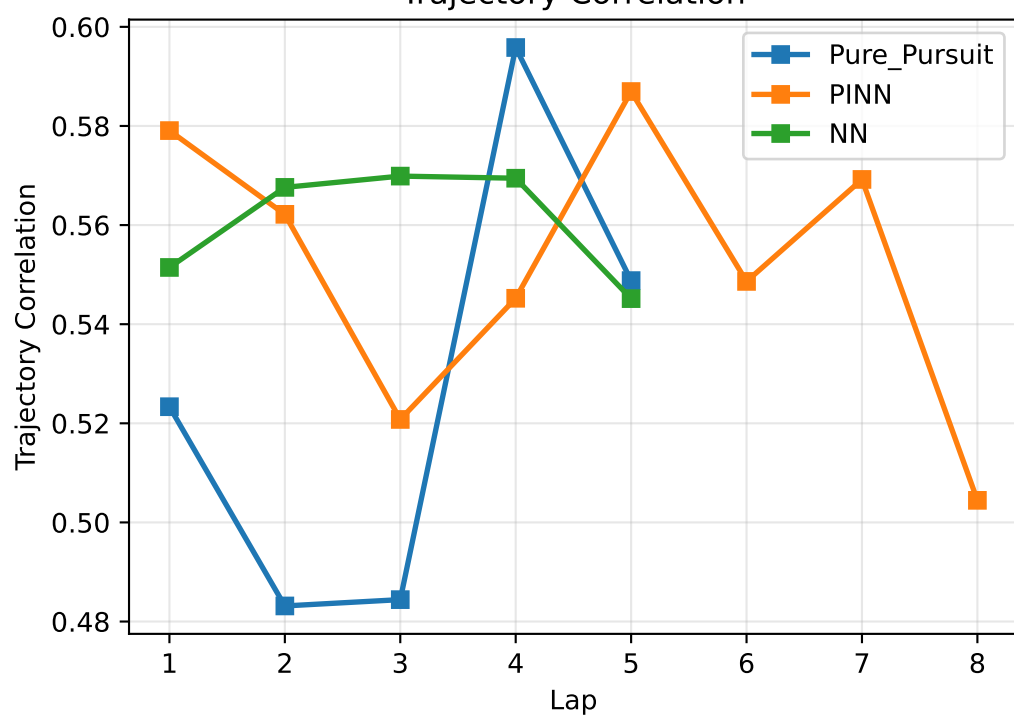
Max CTE (m)



Steering Smoothness (rad/sample)

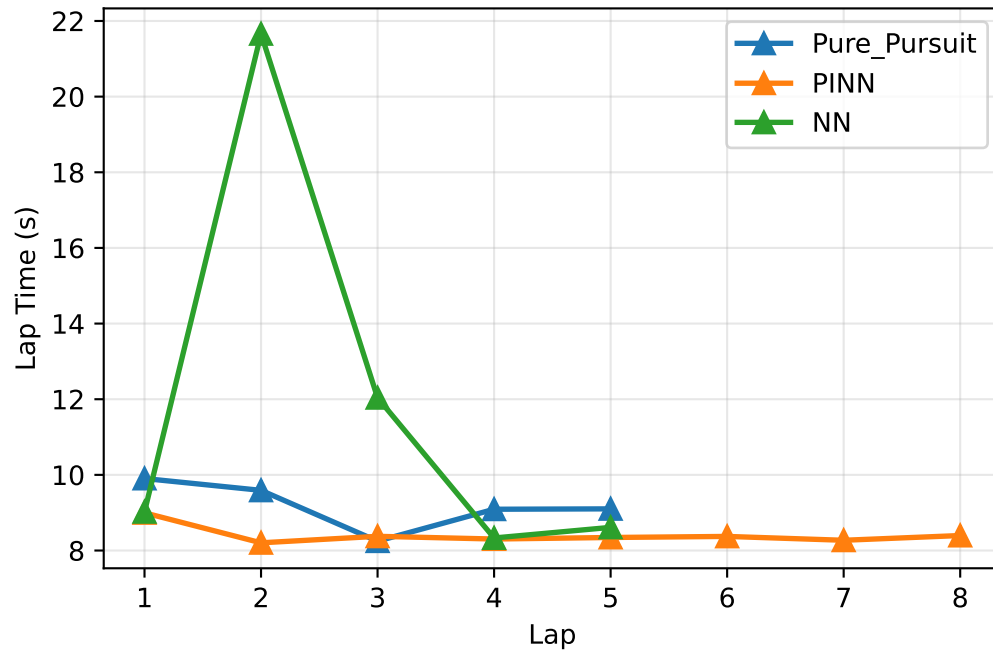


Trajectory Correlation

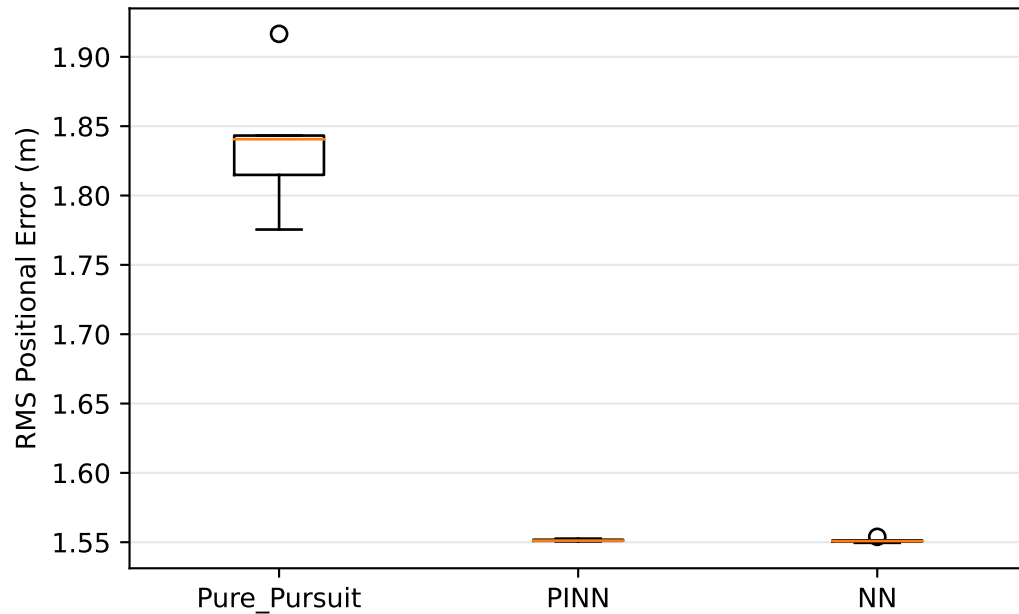


Lap Time & Performance Summary

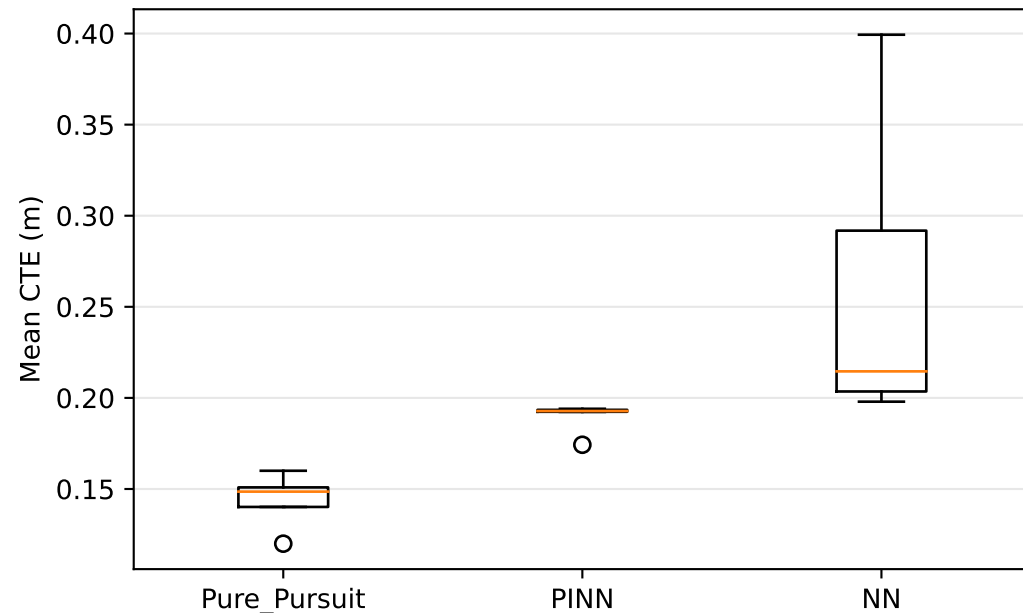
Lap Time Evolution



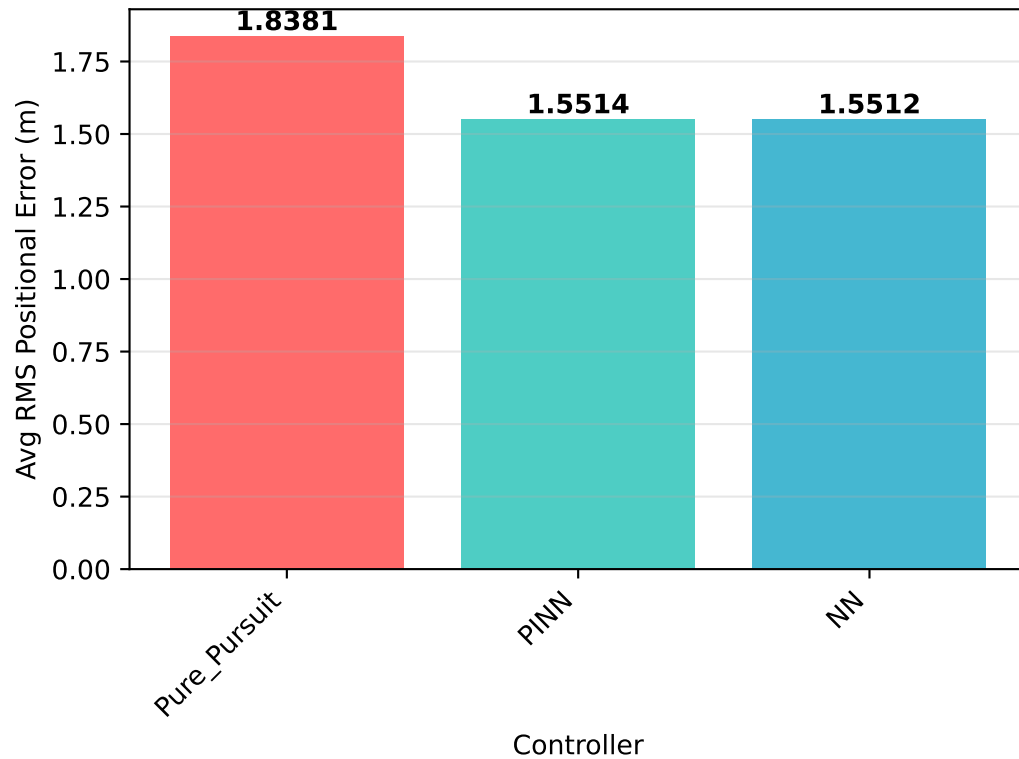
RMS Positional Error Distribution



Cross-Track Error Distribution



Average Performance Comparison



Detailed Per-Lap Metrics

Controller	Lap	Time (s)	RMS X (m)	RMS Y (m)	RMS Pos (m)	RMS CTE (m)	Steer Smooth
Pure_Pursuit	1	9.9037	1.5462	0.9503	1.8149	0.1534	0.3354
Pure_Pursuit	2	9.5907	1.4899	0.9656	1.7755	0.1824	0.3392
Pure_Pursuit	3	8.2478	1.6279	1.0116	1.9166	0.1878	0.3652
Pure_Pursuit	4	9.0913	1.5513	0.9906	1.8406	0.1743	0.3485
Pure_Pursuit	5	9.1015	1.5549	0.9899	1.8432	0.1675	0.3482
PINN	1	8.9961	1.3073	0.8349	1.5511	0.2277	0.3514
PINN	2	8.2036	1.2876	0.8670	1.5523	0.2410	0.3664
PINN	3	8.3757	1.2919	0.8585	1.5512	0.2409	0.3625
PINN	4	8.3038	1.2919	0.8584	1.5511	0.2400	0.3641
PINN	5	8.3445	1.2931	0.8576	1.5516	0.2413	0.3071
PINN	6	8.3729	1.2920	0.8575	1.5506	0.2411	0.3066
PINN	7	8.2709	1.2899	0.8641	1.5526	0.2404	0.3649
PINN	8	8.3962	1.2913	0.8586	1.5507	0.2422	0.3621
NN	1	9.0231	1.3054	0.8368	1.5506	0.2576	0.3511
NN	2	21.6595	1.3969	0.6706	1.5496	0.4427	0.1908
NN	3	12.0384	1.3480	0.7669	1.5508	0.3543	0.3027
NN	4	8.3316	1.2899	0.8664	1.5539	0.2686	0.3072
NN	5	8.6120	1.3060	0.8373	1.5513	0.2576	0.3022